Subramanya M Sagar

Bangalore, India | +91-9986653334 | Subramanya | ashen1queter | subramanyasagar@gmail.com

Education

IDK *Sept 2026 –*
BE in Computer Science

- CGPA: /10.0
- **Coursework:**

Experience

Intern *Bangalore, India*
Codemonk *June 2024 – March 2025*

- Worked on STM32 MCUs, utilizing STM32CubeMX to generate initialization code and configure hardware peripherals for seamless integration.

  Peripherals: USB, UART
- Coded bare-metal applications for keyboards with simple layouts, implementing key matrix scanning, external interrupt (ETI) handling, debounce logic, and power management features like sleep and power-off modes. and communication protocols
- Detailed PCB schematics were created using KiCad, ensuring efficient component placement and robust circuit functionality.
- Custom keyboards built from the ground up by hand-wiring key switches to a microcontroller, focusing on simple and efficient layouts.
- Using Shapr3D to design ergonomic keyboards, focusing on user comfort, optimal placement of mechanical switches, and customizable layouts.

Projects

Breakable Keyboard *github.com/ashen1queter/bRick$_b$Reak*

- Designed a unique keyboard featuring magnetically attachable halves, enabling users to break down the keyboard into four segments for maximum comfort and portability.
- Developed custom firmware from scratch using STM32CubeMX, integrating UART and USB peripherals to enable seamless communication.

  Peripherals: USB, UART
- Language: C
- Tools used: STM32CubeMX, KiCad, Shapr3D

Technologies

Languages: C

Technologies: STM32CubeMX, Shapr3D, KiCad, QMK